THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the data with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 2, 16, 10, 2, 4, 5, 11 respectively, The mode of the data is: 1

(a) 77 **(b)** 70 **(c)** 94.8 **(d)** 80

2. 30 cards numbered 1 to 30 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 or 5 is: 1

(a) $\frac{8}{15}$ **(b)** $\frac{7}{15}$ **(c)** $\frac{13}{30}$ **(d)** $\frac{1}{2}$

3. The least number which when divided by 6, 8 and 12 leaves remainder 3 each time is: 1

(a) 21 **(b)** 27 **(c)** 51 **(d)** 24

4. The value of k for which $x = 2$ is a zero of $x^2 + kx - 10$ is: 1

(a) 6 **(b)** 7 **(c)** -3 **(d)** 3

5. For the pair $kx + 2y = 5$ and $6x + y = 3$ to have a unique solution, k must NOT equal: 1

(a) 8 **(b)** 10 **(c)** 12 **(d)** 3

6. If $x = 4$ is a root of $x^2 - 9x + k = 0$, then $k =$ 1

(a) 20 **(b)** 5 **(c)** 52 **(d)** -20

7. The sum of the first 14 terms of the AP $12, 17, 22, 27, \ldots$ is: 1

(a) 705 **(b)** 546 **(c)** 1246 **(d)** 623

8. A pole 6 m high casts a shadow 4 m long. A tower casting a shadow 24 m has height (m): 1

(a) 144 **(b)** 20 **(c)** 36 **(d)** 26

9. A circle has centre $(1, 2)$ and passes through $(4, 6)$. Its radius is: 1

(a) 5 **(b)** 25 **(c)** 7 **(d)** 1

10. $\dfrac{\sin 20^\circ}{\cos 70^\circ} + \dfrac{\cos 30^\circ}{\sin 60^\circ} =$ 1

(a) 2 **(b)** 0 **(c)** 4 **(d)** 1

11. A storm breaks a tree at a height of 3 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. The distance from the foot of the tree to the point where its top touches the ground is: 1

(a) 6 m **(b)** $3\sqrt{3}$ m **(c)** 3 m **(d)** $\sqrt{3}$ m

12. Two concentric circles have radii 52 cm and 20 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 96 cm **(b)** 64 cm **(c)** 72 cm **(d)** 48 cm

13. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 60° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 154 cm² **(b)** 38.5 cm² **(c)** 462 cm² **(d)** 77 cm²

14. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 3 cm. The length of the wire is: 1

(a) 768 cm **(b)** 128 cm **(c)** 256 cm **(d)** 512 cm

15. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 13, 4, p, 8, 14, 5 respectively is 39.2. The value of p is: 1

(a) 11 **(b)** 5 **(c)** 8 **(d)** 6

16. A bag contains 3 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. The value of x is: 1

(a) 9 **(b)** 5 **(c)** 6 **(d)** 2

17. The decimal expansion of $\frac{77}{210}$ is: 1

(a) non-terminating repeating **(b)** an integer

(c) terminating **(d)** non-terminating non-repeating

18. If the zeroes of $x^2 + px + q$ are 1 and 6, then p equals: 1

(a) -7 **(b)** -6 **(c)** 7 **(d)** 6

19. If $3x + 2y = 25$ and $2x + 3y = 20$, then $x + y =$ 1

(a) $\frac{45}{2}$ **(b)** 5 **(c)** 45 **(d)** 9

20. The equation $x^2 + 6x + 8 = 0$ has: 1

(a) two distinct real roots **(b)** three real roots

(c) no real roots **(d)** two equal real roots

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	6	9	7	5	16	7

Find the mean.

22. A red die and a blue die are rolled together. Find the probability that the sum is a multiple of 3. 2

23. Find the smallest number which leaves remainder 5 when divided by 12, 16 and 18. 2

24. If $x = 3$ is a zero of $x^2 - 8x + k$, find k. 2

25. For what value of k does the pair $kx + 2y = 5$, $6x + y = 3$ have a unique solution? 2

26. If $x = 4$ is a root of $x^2 - 9x + k = 0$, find k and the other root. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. The sum of the digits of a two-digit number is 11. If the digits are interchanged, the number becomes 45 more. Find the number. 3

28. Find the nature of the roots of $3x^2 - 7x + 2 = 0$ and, if real, find them. 3

29. Find the sum of the first 20 terms of the AP whose nth term is $a_n = 3n + 2$. 3

30. $\triangle ABC \sim \triangle PQR$ with $AB = 3$ cm, $BC = 4$ cm, $CA = 5$ cm and $PQ = 6$ cm. Find QR and RP. 3

31. Find the coordinates of the point which divides the join of $(-2,\ 3)$ and $(8,\ 13)$ internally in the ratio $2 : 3$. 3

OR

If $\tan\theta = \dfrac{15}{8}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

32. A building is $6\sqrt{3}$ m high. From the top of the building the angle of elevation of the top of a tower is 30° and from the foot of the building it is 60°. Find the height of the tower and its distance from the building (in this order). 3

OR

A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 18$ cm, $BC = 15$ cm and $CD = 12$ cm. Find the length of AD. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A ladder has 11 rungs whose lengths decrease uniformly from 45 cm at the bottom to 25 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

34. The diagonals of a rhombus are 12 cm and 16 cm. Find its side. 5

35. Find the point on the x-axis which is equidistant from $(1,\ 5)$ and $(7,\ 3)$. 5

36. If $\tan A = \dfrac{15}{8}$, evaluate $\dfrac{3\sin A + 2\cos A}{4\sin A + 1\cos A}$. 5

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 2 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. Find the height of the tower and the initial distance of the point from its foot (in this order). 5

37. Prove that if PA and PB are tangents from P to a circle with centre O, then $\angle APB + \angle AOB = 180^\circ$. 5

OR

An umbrella has 8 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 42 cm. Find the area between two consecutive ribs. 5

38. Water from a full cylindrical tank of radius 7 cm and height 20 cm is poured into an empty rectangular tank of base 22 cm by 14 cm. Find the height of water in the rectangular tank. 5

OR

The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 14, x, 20, y, 19, 3 respectively, and total frequency 100, is 54. Find the value of x. 5

END OF QUESTION PAPER **Total: 80 Marks**

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